Lab 7 - 1. Buggy application

Terminal output

```
$ python3 buggy_processor.py
Input: '1x2\nE123\n{"weight":12.5}\n'
==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): Traceback (most recent call last):
  File "/Users/tanmaysridhar/SE-Labs--PES1UIG25CS849-/Lab 7/buggy_processor.py", line 36, in <module>
    main()
  File "/Users/tanmaysridhar/SE-Labs--PES1UIG25CS849-/Lab 7/buggy_processor.py", line 22, in main
    dim_weight = calculate_dimensional_weight(dims_input)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/tanmaysridhar/SE-Labs--PES1UIG25CS849-/Lab 7/buggy_processor.py", line 7, in calculate_dimensional_weight
    height = float(dims[2])
                   ~~~~^^^
IndexError: list index out of range

Exit code: 1
```

Lab 7 - 2. Failing tests

Terminal output

```
$ python3 -m pytest -q --tb=no test_processor.py
FFF...F.F.FFFFF.FFFFF..FFFFFF                                            [100%]
========================== short test summary info ============================
FAILED test_processor.py::test_calculate_dimensional_weight_no_crash - Except...
FAILED test_processor.py::test_format_tracking_label_safe - ExceptionGroup: H...
FAILED test_processor.py::test_parse_package_manifest_safe - ExceptionGroup: ...
FAILED test_processor.py::test_invalid_dimensions[None] - AttributeError: 'No...
FAILED test_processor.py::test_invalid_dimensions[1x2] - IndexError: list ind...
FAILED test_processor.py::test_invalid_dimensions[0x2x3] - Failed: DID NOT RA...
FAILED test_processor.py::test_invalid_dimensions[nanx2x3] - Failed: DID NOT ...
FAILED test_processor.py::test_invalid_dimensions[1e308x1e308x1e308] - Failed...
FAILED test_processor.py::test_invalid_tracking[None] - TypeError: 'NoneType'...
FAILED test_processor.py::test_invalid_tracking[] - IndexError: string index ...
FAILED test_processor.py::test_invalid_tracking[123] - Failed: DID NOT RAISE ...
FAILED test_processor.py::test_invalid_tracking[E-123] - Failed: DID NOT RAIS...
FAILED test_processor.py::test_invalid_tracking[E1234567] - Failed: DID NOT R...
FAILED test_processor.py::test_invalid_tracking[A\uff11\uff12] - Failed: DID ...
FAILED test_processor.py::test_invalid_manifest[None] - TypeError: the JSON o...
FAILED test_processor.py::test_invalid_manifest[[]] - TypeError: list indices...
FAILED test_processor.py::test_invalid_manifest[{}] - KeyError: 'weight'
FAILED test_processor.py::test_invalid_manifest[{"weight":null}] - TypeError:...
FAILED test_processor.py::test_invalid_manifest[{"weight":true}] - Failed: DI...
FAILED test_processor.py::test_invalid_manifest[{"weight":-1}] - Failed: DID ...
FAILED test_processor.py::test_invalid_manifest[{"weight":1e999}] - Failed: D...
21 failed, 8 passed in 0.34s

Exit code: 1
```

Lab 7 - 3. Fixed application

Terminal output

```
$ python3 processor.py
Input: '1x2\nE123\n{"weight":12.5}\n'

==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): Input error: Expected exactly three dimensions (LxWxH)

==== Running format_tracking_label ====
Enter raw code (e.g. E123): Result: E-000123

==== Running parse_package_manifest ====
Enter manifest JSON (e.g. {"weight": 12.5}): Result: 12.5

Exit code: 0
```

Lab 7 - 4. Passing tests

Terminal output

```
$ python3 -m pytest -q test_processor.py
.............................                                        [100%]
29 passed in 0.45s

Exit code: 0
```